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Section: 5A

Course: ADC

Question # 01

$$g(t) = e^{-4t}u(t)$$

$$e^{-4t}u(t) \Longleftrightarrow \frac{1}{4+jw}$$

(a) $\Psi_g(w) = ?$

$$\Psi_g(w) = |h(w)|^2$$

$$= \left(\frac{1}{4+jw}\right)^2$$

$$\Psi_g(w) = \frac{1}{16+w^2}$$

(c) $E_{out} = 0.9\,E_g$

$$E_y = 0.9 \times \frac{1}{8}$$

$$E_y = 0.1125$$

$$\frac{1}{\pi}\tan^{-1}\left(\frac{w_c}{4}\right) = 0.1125$$

$$\frac{w_c}{4} = \tan(0.1125\pi)$$

$$\boxed{w_c = 1.476 \text{ rad/s}}$$

$$H(w) = \begin{cases} 2 & |w| \le w_c \\ 0 & |w| > w_c \end{cases}$$

(b) $E_g = ?$

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2\,dt$$

$$E_g = \int_{0}^{\infty} e^{-4t}\,dt$$

$$E_g = \frac{e^{-8t}}{-8}\Big|_0^{\infty} = 0 + \frac{1}{8} = \frac{1}{8}\ J$$

$$E_y = \frac{1}{2\pi}\int_{-\infty}^{\infty} \Psi_g(w)\,dw$$

$$E_y = \frac{1}{2\pi}\int_{-\infty}^{\infty} |H(w)|^2\,\Psi_g(w)\,dw$$

$$E_y = \frac{1}{2\pi}\int_{-w_c}^{w_c} (2)^2 \frac{1}{16+w^2}\,dw$$

$$E_y = \frac{4}{2\pi}\int_{-w_c}^{w_c} \frac{1}{16+w^2}\,dw$$

$$= \frac{2}{\pi} \times \frac{1}{4}\tan^{-1}\left(\frac{w_c}{4}\right)\Big|_{-w_c}^{w_c}$$

$$= \frac{1}{\pi}\tan^{-1}\left(\frac{w_c}{4}\right)$$

$$= \frac{1}{\pi}\tan^{-1}\left(\frac{w_c}{4}\right)$$

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Question # 02

$$y(t) = g(t) + \alpha g(t-\tau) \qquad \alpha = 0.2, \quad \tau = 5\,\mu s$$

(a)

$$Y(\omega) = h(\omega) + \alpha h(\omega) e^{-j\omega\tau}$$

$$Y(\omega) = h(\omega)\left[1 + \alpha e^{-j\omega\tau}\right]$$

$$\frac{Y(\omega)}{h(\omega)} = 1 + \alpha e^{-j\omega\tau}$$

$$H(\omega) = 1 + 0.2 e^{-j\omega 5\mu}$$

(b)

$$H(\omega) = 1 + 0.2\left[\cos(5\mu\omega) - j\sin(5\mu\omega)\right]$$

$$H(\omega) = 1 + 0.2\cos(5\mu\omega) - j0.2\sin(5\mu\omega)$$

$$|H(\omega)| = \sqrt{\left[1 + 0.2\cos(5\mu\omega)\right]^2 + \left[0.2\sin(5\mu\omega)\right]^2}$$

$$= \sqrt{1 + 0.4\cos 5\mu\omega + 0.04\cos^2(5\mu\omega) + 0.04\sin^2(5\mu\omega)}$$

$$= \sqrt{1 + 0.4\cos 5\mu\omega + 0.04\left[\cos^2(5\mu\omega) + \sin^2(5\mu\omega)\right]}$$

$$= \sqrt{1 + 0.4\cos 5\mu\omega + 0.04}$$

$$|H(\omega)| = \sqrt{1.04 + 0.4\cos 5\mu\omega}$$

$$\theta_n(\omega) = \tan^{-1}\left(\frac{-0.2\sin 5\mu\omega}{1 + 0.2\cos 5\mu\omega}\right)$$

(c) For a channel to be distortionless, its amplitude response must be perfectly flat (constant) across all frequencies, and its phase response must be perfectly

linear function of ω. Because $|H(\omega)| = \sqrt{1.04 + 0.4\cos(\omega\tau)}$ fluctuates sinusoidally with frequency, this channel causes Amplitude distortion. Additionally, trigonometric ratio involved, it is non-linear, causing phase delay.

Set the cos argument to the first angle that yields -1 (which is π rad).

$$\cos(\omega\tau) = -1 \Rightarrow \omega\tau = \pi$$

$$2\pi f \tau = \pi$$

$$f = \frac{\pi}{2\pi \times 5\mu}$$

$$\boxed{f = 100\ kHz}$$

Question # 03

$$g(t) = 4\cos(100t) + 2\sin(300t)$$

~~$G(\omega) = 4\pi[\delta(\omega-100) + \delta(\omega+100)] + \frac{2\pi}{j}[\delta(\omega-300) - \delta(\omega+300)]$~~

For any pure sinusoid $x(t) = A\cos(\omega_0 t + \theta)$ or $A\sin(\omega_0 t + \theta)$

$$S_x(\omega) = \frac{\pi A^2}{2}[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$$

$$S_x(\omega) = \frac{4^2\pi}{2}[\delta(\omega-100) + \delta(\omega+100)] + \frac{2^2\pi}{2}[\delta(\omega-300) + \delta(\omega+300)]$$

$$S_x(\omega) = 8\pi[\delta(\omega-100) + \delta(\omega+100)] + \frac{2\pi}{1}[\delta(\omega-300) + \delta(\omega+300)]$$

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(b) As $P = \frac{A^2}{2}$

$$P_i = \frac{(4)^2}{2} + \frac{(2)^2}{2}$$

$$P_i = 8 + 2 = 10$$

(c)

$$H(\omega) = \frac{1}{1 + j\frac{\omega}{200}}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{200}\right)^2}}$$

For $\omega = 100$ $P_i = 8$

$$P_{out} = P_i \times |H(\omega)|$$

$$= 8 \times \frac{1}{\sqrt{1 + \left(\frac{100}{200}\right)^2}}$$

$$P_{out} = 6.4\,W$$

For $\omega = 300$ $P_i = 2$

$$P_{out} = P_i \times |H(\omega)|$$

$$= 2 \times \frac{1}{\sqrt{1 + \left(\frac{300}{200}\right)^2}}$$

$$P_{out} = 0.615\,W$$

$$P_{T,out} = 6.4 + 0.615$$

$$\boxed{P_{T,out} = 7.015\,W}$$

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Question # 04

(a) For distortionless transmission. B.P Lathi states the output must be a scaled, time-delayed exact replica of the input: $y(t) = A\,x(t-t_d)$

This dictates a transfer function of $H(\omega)$ $Ae^{-j\omega t_d}$ which enforces two strict criteria ove the signal bandwidth.

$$|H(\omega)| = A$$

$$\theta_n(\omega) = -\omega t_d$$

(b)

Phase delay (t_p):-

$$t_p(\omega) = \frac{-\theta_n(\omega)}{\omega}$$

$$= \frac{-(-\omega t_0 - \beta\omega^3)}{\omega}$$

$$t_p(\omega) = t_0' + \beta\omega^2$$

Group delay (t_g):-

$$t_g(\omega) = -\frac{d\,\theta_n(\omega)}{d\omega} \Rightarrow \frac{d}{d\omega}(\omega t_0 + \beta\omega^3)$$

$$t_g(\omega) = t_0 + 3\beta\omega^2$$

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(c) Envelope distortion:-

Yes (specifically, dispersion), Because the group delay $t_g(\omega) = t_o + 3B\omega^2$ is frequency-dependent, different frequency components of the pulse experience different delays. This causes the wave packet to spread out over time.

Physical Meaning:-

Group delay is the actual time it takes for the signal's energy (the envelop) to travel through the channel. In contrast, phase delay is merely the travel time of the underlying carrier wave.

Question # 05

$B = 5\,kHz$

$S_n(\omega) = \frac{N_0}{2} = 10^{-8}\ W/rad/s$

$SNR = 35\ dB$

$SNR_{loss} = 20\ dB$

(a) $SNR_{dB} = 10\log_{10}(SNR_{linear})$

$35 = 10\log_{10}(SNR_{linear})$

$10^{3.5} = SNR_{linear}$

$SNR_{linear} = 3162.28$

(b) Finding total Noise N

$\omega_B = 2\pi(5000) = 10000\ rad/s$

$N = \frac{1}{2\pi}\int_{10,000}^{10000} S_n(\omega)\,d\omega$

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$$N = \frac{1}{2\pi}\int_{-10{,}000\pi}^{10{,}000\pi} 10^{-8}\, d\omega$$

$$= \frac{10^{-8}}{2\pi}\,\omega\Big|_{-10{,}000\pi}^{10{,}000\pi}$$

$$N = 10^{-4}\omega$$

$$SNR_{linear} = \frac{P_{rx}}{N}$$

$$P_{rx} = 10^{3.5} \times 10^{-4}$$

$$\boxed{P_{rx} = 0.3162\omega}$$

(c)

$$SNR_{loss} = 10\log_{10}(L)$$

$$20 = 10\log_{10}(L)$$

$$L = 100$$

$$P_{tx} = P_{rx} \times L$$

$$= 10^{-0.5} \times 100$$

$$= 10^{1.5}$$

$$P_{tx} = 31.62\omega$$

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